

CECILIA NYBERG

M.Sc. Data Science and AI

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WORK EXPERIENCE

Chief Product Officer & AI Systems Engineer

LLM Core AI (Startup), Seoul National University

📅 August 2025 – Present 📍 Seoul, South Korea
Product design and technical development of an LLM agent for Korean skincare e-commerce. Implemented LangGraph-based orchestration for routing, RAG, memory, and tool calls, and built API communication with a Next.js interface and a PostgreSQL/Supabase backend. Startup webpage: <https://llmcore.ai/en>

Research Assistant

Ocean Engineering Deep Learning Lab, Seoul National University

📅 July 2025 – Present 📍 Seoul, South Korea
Conducting research on LLM inference, cascade networks, and tool learning with RLHF.

Master's Thesis Intern

Ericsson

📅 January 2025 – June 2025 📍 Stockholm, Sweden
Designed a dynamic voltage switching system for 4G radio units by forecasting traffic load using PRB utilization and real-time power data; evaluated transformer-based and classical ML models; and analyzed interpretability using SHAP.

Software Developer

Ongoing WMS: Warehouse Management System

📅 June 2024 – July 2024 📍 Gothenburg, Sweden
Implemented service selection functionality for delivery bookings, working across front- and back-end using C#, HTML, and SQL.

Product Strategy Intern

Firework

📅 June 2023 – August 2023 📍 San Mateo, California
Developed a B2B e-commerce launch strategy through market research, competitor analysis, and customer segmentation.

CURRENT PROJECTS

Knowledge Distillation for Early Exit Systems

Chalmers University of Technology

📅 August 2025 – Present
Investigating how knowledge distillation can improve the training of early-exit neural networks for energy-efficient AI. Comparing strategies for effective distillation design to maintain accuracy while reducing inference cost.

Tool Learning with Prompt Augmentation

Seoul National University

📅 September 2025 – Present
Developing methods to improve LLMs' tool-using abilities in the ship inspection domain through mixed prompt SFT, aiming to address reliability and correctness challenges in tool use for smaller models.

For more projects, see portfolio: <https://cecnyb.github.io/>

COMPETENCES

Machine Learning

LLM Agents

Transformers

RLHF

RAG

Python

Linux

PyTorch

LangGraph

Ollama

FastAPI

SSE Streaming

Quantization

LANGUAGES

Swedish

English

Spanish

German

Korean

EDUCATION

M.Sc. Data Science and AI,
GPA: 4.88/5

Chalmers University of Technology

📅 September 2023 – June 2025

Exchange semester, Data
Science and AI, GPA: 95%

Seoul National University

📅 September 2024 – December 2024

Summer Session

University of California, Berkeley

📅 June 2023 – August 2023

B.Sc. Industrial
Engineering and
Management, GPA: 4.78/5

Chalmers University of Technology

📅 September 2020 – June 2023

SELECTED PAST PROJECTS

Accessible Game Application for the Visually Impaired

Bachelor's Thesis, Chalmers University of Technology

📅 January 2023 – June 2023

Designed and developed an AI-driven mobile application to represent card hands through audio and tactile feedback, enabling inclusive board game play for users with visual impairments.

Meta-Learning for Transferability of WACO: Workload-Aware Co-Optimization for Sparse Tensor Programs

Seoul National University

📅 October 2024 – December 2024

Integrated meta-learning into WACO's training loop to improve cross-architecture generalization in sparse matrix operations. Reduced fine-tuning data requirements and mitigated WACO's high adaptation cost on new hardware.

For more projects, see portfolio: <https://cecnyb.github.io/>